

Muhammad Yasin

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PROFESSIONAL SUMMARY

Full-stack software engineer specializing in Django/DRF, React, and Next.js, with a track record of designing scalable backend systems and AWS-based infrastructure. Worked across multiple backend frameworks, including FastAPI and Express.js, with experience in RESTful API design, system architecture, and shipping production-ready applications with integrated AI capabilities.

PROFESSIONAL EXPERIENCE

Software Engineer

Innovazel Private Limited

Aug 2025 – Present

Lahore, Punjab, Pakistan

Core Technologies: Django, DRF, React, PostgreSQL, Celery, Redis, AWS, GitHub Actions

• PeopleX HRM — Internal SaaS HR Management Product

- Serving as main developer, alongside the engineering lead, on PeopleX, an internal HRM SaaS product built with Django REST Framework and PostgreSQL that the company is developing for commercial sale, consolidating employee management, leave & attendance, CRM, and finance operations into a unified platform with 100+ RESTful API endpoints.
- Built a configurable leave policy engine supporting per-type rules including probation eligibility, carry-forward limits, advance notice, and maximum consecutive days, with automated nightly balance recalculation via Celery.
- Developed a custom in-house e-signature workflow supporting multi-recipient sequential signing with a full audit trail (IP, user agent, timestamps), removing dependency on third-party providers like DocuSign.
- Implemented JWT and Google OAuth 2.0 based authentication with role-based access control, alongside Google Workspace account provisioning to automate new-hire onboarding.
- **Advance Alloy Wheel Repairs — Business Management Platform** (advancedalloys.com.au)
- Built the platform's foundational Django/DRF API endpoints and PostgreSQL schemas during its early phase, then grew it into a full production-grade business management system for a high-profile Australian automotive client, replacing WorkflowMax, HubSpot CRM, and Asana with one unified platform.
- Architected a scalable backend with Redis-backed Celery workers and a cron job scheduler to handle background processing across data-sync pipelines, scheduled invoicing jobs, and high-volume third-party reconciliation workflows.
- Reduced manual invoicing effort by approximately 80% by integrating Xero's financial APIs to automate the full invoice lifecycle, including tax-rate handling and a three-layer customer-resolution process, and backfilled 37,000+ historical invoices from Xero into the platform.
- Removed polling overhead by implementing Xero webhook listeners to process financial events in real time, keeping accounting state synchronized across the platform with minimal latency.
- Built a responsive React.js frontend using short polling to maintain live UI state across data-heavy financial and operational modules used daily in production.
- Owned the DevOps pipeline end-to-end — AWS S3 for media storage, Nginx for static/media serving, isolated EC2 instances for staging and production, and GitHub Actions CI/CD secured with GitHub Secrets — enabling zero-downtime deployments across environments.
- **360 HR — Client HR Management Platform**
- Contributed to backend development of a client-facing HR management platform, building API functionality to support employee and organizational data management.
- Collaborated with the team to translate client requirements into scalable backend features within the MERN(Mongodb, Express.js, React.js, Node.js) stack.

Software Engineer Intern

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Mar 2025 – Jul 2025

Lahore, Pakistan

Core Technologies: React.js, Node.js, Express.js, MongoDB

- Built and maintained UI components and pages in React.js as part of a MERN-stack development team, contributing to multiple internal features across different client-facing modules.
- Developed backend REST API endpoints using Node.js and Express.js to support frontend functionality and data flow between client and server.
- Designed and worked with MongoDB schemas and queries to support application data storage and retrieval needs.
- Identified and resolved bugs across the frontend and backend, improving feature stability and code quality under guidance from senior engineers.

EDUCATION

Bachelor of Science in Computer Science <i>Bahria University</i>	CGPA 3.83/4.0 <i>Gold Medalist</i>
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PROJECTS

DiagnoSense — AI Lung Cancer Detection Platform (FYP) <i>React, Node.js, FastAPI, MongoDB</i> • Built a full-stack MERN application integrating a hybrid CNN and Vision Transformer model to detect early-stage lung cancer from CT scan images, with predictions served via a dedicated FastAPI backend. • Implemented secure JWT-based authentication, user dashboards, and route handling in Express.js, connecting the ML inference layer to a responsive React frontend. • Integrated a RAG-based chatbot to deliver AI-driven health insights to users, extending the platform beyond diagnosis into interactive patient support. • Designed the system architecture end-to-end across frontend, backend, and ML services, prioritizing usability and real-world clinical relevance.
Brain Tumor MRI Classification — Hybrid CNN-Transformer Diagnostic Web App <i>FastAPI, TensorFlow/Keras, OpenCV, JavaScript</i> • Built an end-to-end web application serving an EffViT-Hybrid model (EfficientNetB0 backbone with a ViT-Tiny style transformer head) to classify brain MRI scans into glioma, meningioma, pituitary tumor, and no-tumor categories. • Designed a weights-only model loading strategy that rebuilds the architecture in code rather than deserializing the full saved model, avoiding cross-version Lambda-layer failures and improving deployment portability. • Implemented custom Keras layers (transformer block, positional embedding with CLS token, CLS token extraction) to support safe model reconstruction and reuse across environments. • Built a Grad-CAM visualization pipeline tailored to the hybrid architecture, forwarding CNN stem activations through the transformer head to generate spatial attention overlays, with non-blocking fallback if visualization fails. • Delivered a FastAPI backend with upload validation, a JSON prediction API, and a health-check endpoint, paired with a responsive frontend for uploading scans and viewing predictions, confidence scores, and attention overlays.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, SQL Backend: Django, Django REST Framework, FastAPI, Node.js, Express.js Frontend: React.js, Next.js Databases: PostgreSQL, MongoDB DevOps & Cloud: AWS (EC2, S3, RDS), Docker, Nginx, GitHub Actions, Celery AI/ML: CNN-based Models, Vision Transformers, Retrieval-Augmented Generation (RAG) Tools: Git, JWT Authentication, Oauth, RESTful APIs

ADDITIONAL INFORMATION

Languages: English (Fluent), Urdu (Native)
